From Mills Clocks and Euler's Constant to Ten External Problems

Complete Answers, Exact Identities, and Partial Arithmetic Theorems

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A single research manuscript with full proofs and a status register

Abstract

We apply modular return, integer-tail, monodromy, and residue methods from a preceding investigation of Euler's constant and shifted Mills constants to ten externally formulated targets. Two questions of Saito are answered with their full quantifiers by reconstructing earlier corpus arguments: a non-reversible inhomogeneous recurrence produces composite Pisot-floor windows, and every fixed integer shift of the dyadically indexed Fibonacci sequence is composite on a positive-density set of indices. We also resolve the analytic-identity component of a recent Meijer-function extension problem: explicit differential operators evaluate to 1 and $\gamma + \operatorname{Log} z$ at every argument, with a higher zeta-polynomial hierarchy and an exact telescoping kernel. Further theorems classify E-value units in the field of all algebraic exponential values with one Ein value, prove irrationality for power-growth subfamilies of the monotone Erdős 270 problem, classify primitive harmonic-lattice seeds, and give an all-rank scalar approximation to $\gamma + \log z$. Three additional prime-clock targets receive precise partial results. Every external target is matched to its original source and its actual outcome. Seven of the ten targets remain unresolved in their full stated scope. Neither irrationality of γ nor unconditional transcendence of the classical Mills constant is proved.

Keywords. Mills constants; Euler–Mascheroni constant; Pisot numbers; Lucas sequences; E-function values; monodromy; factorial series; Meijer G-functions; conservative matrix fields.

Reading convention. A complete proof in this manuscript is distinct from a claim of historical priority or external peer review. The source register identifies prior corpus arguments, new derivations in this investigation, and remaining hypotheses.

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1 The ten targets and their exact outcomes

This paper follows [17], which organized ten gamma-related and ten Mills-related items from the research dialogue. The present task is different: select ten actual questions outside that discussion and determine what the developed methods prove about them. We keep each external question's quantifiers visible. An auxiliary irrational series is not a proof that its base is irrational; a classification on one field is not a classification of every E-value; and scalar convergence is not generic matrix-state convergence.

Our notation $\xi((C_n))$ means the least $A > 1$ for which $\lfloor A^{C_n} \rfloor$ is prime for every $n \geq 1$, when that least value exists. In particular

$$M = \xi((3^n)), \qquad \xi_2 = \xi((2^n)).$$

The four shifted clocks $3^n \pm 1$ and $3^n \pm n$ motivated the preceding investigation. The full external answers below use their modular-clock and arithmetic methods; they do not require an unproved identity expressing γ in terms of those four constants.

1.1 Status and provenance

The register uses three outcomes. *Full answer* means all quantifiers in the named external problem are proved here. *Identity component resolved* means the precisely selected analytic component of a broader future-work paragraph is proved. *Partial; original target open here* means a complete auxiliary theorem is provided, but the original external conclusion is not. No statement that a source problem was globally unsolved immediately before this manuscript is inferred merely from an unsuccessful search. Sources and versions were checked on 5 November 2026.

The two full Saito answers already existed in corpus manuscripts [007] and [008], dated 5 September 2026; we reconstruct their proofs and strengthen the exposition rather than count them as discoveries first made today. The conditional Mills-gap and three-adic-window arguments likewise have identified corpus predecessors. The residue, field-intersection, denominator-window, and scalar-concentration derivations are developed in the present investigation; literature priority for equivalent results is not asserted.

ID	External target and source	Outcome in this manuscript
T1	Irrationality of Euler's constant γ; current approximation context [16].	**Partial.** Exact primitive-seed classification, height cost, and a conditional radical-membership obstruction. The irrationality question is not closed.
T2	Classical Mills transcendence, through Saito's cubic-Pisot alternative, Question 1.7 [1].	**Partial.** A hypothetical algebraic tower has a transcendental gap exponent and divisibility constraints. The cubic-Pisot alternative is not excluded.
T3	Is the least binary-clock constant ξ_2 irrational? Question 1.6 [1].	**Partial.** The reciprocal floor series is irrational for every real base $A > 1$. This does not decide the arithmetic of ξ_2.
T4	Infinitely many composite floors on 3^n for every cubic Pisot number; Problem 1.1 [3].	**Partial.** Nonintegral three-adic trace limits give simultaneous composite windows. Exceptional integral limits remain.
T5	Find a non-reversible ILRS working for every Pisot number; Problem 1.7 [3].	**Full answer, reconstructed from [008].** $R(n) = ba^n \pm n$ works, with simultaneous windows and an extension to Salem numbers.
T6	Infinitely many composite $F_{2^n} + h$ for every integer h; Problem 1.8 [3].	**Full answer, reconstructed from [007].** A positive lower density of composite indices, with a general Lucas-sequence version.

ID	External target and source	Outcome in this manuscript
T7	Are all units of the E-value ring αe^{β}? Conjecture 1.2 [9].	**Partial.** Complete classification over the field of all algebraic exponential values and one nonzero algebraic Ein argument; the global conjecture remains.
T8	Irrationality of the factorial series for every nondecreasing $f \to \infty$; monotone Erdős 270 variant [14].	**Partial.** All $f(n) = \lfloor cn^{\alpha} \rfloor$, $c > 0$, $\alpha > 3/2$, and a boundary range; all integer-valued polynomials of degree at least two.
T9	Extend the Meijer identity to $\gamma + \log z$; analytic component of Section 9.2(1) [16].	**Identity component resolved.** Every $p \geq 2$ and principal-branch z, a zeta-polynomial hierarchy, and the exact kernel of weighted telescoping.
T10	Generic dual convergence for all admissible ranks and trajectories; Conjecture 1 [16].	**Partial.** A separate all-rank scalar convergence theorem and integer dilations of the known matrix family. General dual-state isolation is not proved.

Problem and theorem numbers for Saito refer to the arXiv versions specified in the bibliography; journal versions need not use the same numbering. The unproved Riemann Hypothesis is not inserted as an assumption into an allegedly unconditional conclusion. The implication from the Density Hypothesis to Mills transcendence is an existing external theorem [2].

1.2 The common proof mechanisms

There are three useful transfers. On prime clocks, an accessible residue class turns a single fixed divisor into infinitely many composite values. In rational series, a common denominator turns a nonzero small tail into an impossible integer. For E-values, analytic continuation produces algebraic embeddings whose incompatible ramification and denominator patterns force descent. The Meijer calculation uses a different exact mechanism: an infinite residue sum telescopes to one boundary residue. Each method proves a finite, checkable implication; the remaining challenge is to establish the hypothesis needed by the original target.

2 Euler's constant: exact arithmetic of adaptive seeds

The first target is the irrationality of

$$\gamma = \lim_{n\to\infty}(H_n - \log n), \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}.$$

The current recurrence construction of [16] does not settle this question. This section gives an exact arithmetic classification of the two-dimensional image lattice appearing in the preceding integrated manuscript [17]. Its purpose is to test whether selecting a new seed at every depth creates an irrationality proof. The classification shows exactly what this selection costs.

2.1 Primitive points on a harmonic shear lattice

For integers $D \geq 1$ and C with $\gcd(C, D) = 1$, define

$$A_{D,C} = \begin{pmatrix} D & C \\ 0 & D \end{pmatrix}, \qquad \Lambda_{D,C} = A_{D,C}\mathbb{Z}^2.$$

The normalized image in the rank-three trajectory examined in [17, Section 8] has this form with $D = \mathrm{den}(3H_n)$ and $C = 3DH_n$. The theorem below is independent of that identification: it holds for every displayed integral shear, and its proof needs no analytic estimate for a matrix recurrence.

Theorem 2.1 (Exact primitive-seed classification)**.** *Let* $(p, q) \in \mathbb{Z}^2$ *satisfy* $q > 0$ *and* $\gcd(p, q) = 1$*. Put*

$$h = \gcd(Dp - Cq, Dq).$$

Then $h \mid D^2$*, and the unique primitive integer input whose image is a positive multiple of* (p, q) *is*

$$(a, b) = \left(\frac{Dp - Cq}{h}, \frac{Dq}{h} \right). \tag{2.1}$$

Its image and the gcd of the two image coordinates are

$$A_{D,C}(a, b)^T = \frac{D^2}{h}(p, q)^T, \qquad \gcd(Da + Cb, Db) = \frac{D^2}{h}. \tag{2.2}$$

If $\gcd(q, D) = 1$*, then* $h = 1$ *and this output gcd is exactly* D^2*.*

Proof. The defining gcd makes (a, b) integral and primitive. Also h divides

$$D(Dp - Cq) + C(Dq) = D^2 p, \qquad D(Dq) = D^2 q.$$

Bezout's identity for p, q therefore gives $h \mid D^2$. Matrix multiplication proves (2.2); primitiveness of (p, q) identifies its gcd. Any input on the same image ray lies on the rational line spanned by $(Dp - Cq, Dq)$. The only primitive integer point on its positive ray is the one in (2.1).

Finally suppose $\gcd(q, D) = 1$ and a prime ℓ divides h. If $\ell \mid q$, then $\ell \nmid Dp$, contradicting $\ell \mid Dp - Cq$. Otherwise $\ell \mid D$, since $\ell \mid Dq$, and then $\ell \nmid Cq$, giving the same contradiction. Thus $h = 1$. □

Every rational slope is consequently represented. This is an exact surjectivity statement about slopes, not an estimate on the quality of approximation to γ.

Corollary 2.2 (The cost of a chosen rational approximation)**.** *Suppose* $D_n = \mathrm{den}(3H_n)$*,* $C_n = 3D_nH_n$*, and* p_n/q_n *is a bounded sequence of reduced rational numbers with* $q_n > 0$*. With* h_n *as in Theorem 2.1, the effective primitive input satisfies*

$$\max(|a_n|, |b_n|) \sim \frac{3D_n q_n \log n}{h_n}.$$

The image linear form and its primitive normalization are exactly

$$(D_n b_n)\gamma - (D_n a_n + C_n b_n) = \frac{D_n^2}{h_n}(q_n\gamma - p_n),$$

and $q_n\gamma - p_n$*, respectively.*

Proof. Equation (2.1) gives $a_n = (D_n q_n/h_n)(p_n/q_n - 3H_n)$ and $b_n = D_n q_n/h_n$. Since $H_n/\log n \to 1$, the first coordinate eventually dominates and has the claimed size. The linear-form identity follows by substitution in (2.2). □

Here "effective input" means the two free coordinates of the image lattice. The norm of a three-coordinate preimage under the original CMF matrix is a further question. A bound for a fixed matrix seed is not automatically uniform for the depth-dependent inputs just constructed.

2.2 What an irrationality transfer must still prove

Proposition 2.3 (Primitive integer-form criterion)**.** *If integers p_j, q_j satisfy*

$$0 < |q_j\gamma - p_j| \longrightarrow 0,$$

then γ is irrational. Obtaining only $p_j/q_j \to \gamma$ does not supply this conclusion.

Proof. If $\gamma = u/v$ with $v > 0$, each nonzero $q_j\gamma - p_j$ has absolute value at least $1/v$. For the final distinction, even a rational number x has rational approximations $p_j/q_j \to x$; division by q_j can hide the nonvanishing size of its integer linear form. □

The seed classification neither proves the strict nonvanishing nor the decay required by this proposition. It shows that arbitrary adaptive seed selection simply re-encodes the selected rational approximation, with its exact arithmetic multiplier. The decisive remaining estimate must concern the primitive form after this multiplier has been removed.

2.3 A second, algebraic transfer obstruction

Let $\mathcal{E}$ be the ring of E-values, $\mathcal{K}$ its fraction field, and put

$$\eta = \mathrm{Ein}(1), \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\,dt.$$

The usual exponential-integral identity [13, Section 6.2] gives

$$\eta = \gamma + \delta/e. \tag{2.3}$$

The classification in Theorem 4.3, proved below, turns this relation into the following exact conditional obstruction.

Corollary 2.4 (A radical membership obstruction)**.** *If γ is algebraic, then*

$$\delta \in \mathcal{E} \setminus \mathcal{E}^\times, \qquad \delta^{-1} \notin \mathcal{E}, \qquad \delta^{1/m} \notin \mathcal{K} \quad (m \geq 2).$$

Here the last assertion holds for every choice of an mth root. Consequently, proving $\delta^{1/m} \in \mathcal{K}$ for even one $m \geq 2$ would prove that γ is transcendental.

Proof. Write $A = \overline{\mathbb{Q}}$, $\mathcal{B} = A[e^\beta : \beta \in A]$, $\mathcal{L} = \mathrm{Frac}(\mathcal{B})$, and $F = \mathcal{L}(\eta)$. Under the displayed hypothesis, $\delta = e(\eta - \gamma)$ is a nonconstant polynomial in η over $\mathcal{B}$. Theorem 4.3 makes it a non-unit of $\mathcal{E}$. The same theorem says that F is relatively algebraically closed in $\mathcal{K}$. An mth root of δ in $\mathcal{K}$ would therefore lie in F. In the rational function field $\mathcal{L}(\eta)$, the valuation at the prime $\eta - \gamma$ of δ equals one. The valuation of an mth power is divisible by m, a contradiction. □

No membership statement for such a radical is established here. The corollary specifies a possible transfer theorem; it is not a proof that its antecedent can be met. In particular, the irrationality of γ remains unresolved in this manuscript.

3 Five external questions about prime-representing constants

The five targets below retain the quantifiers of their published formulations. Two are answered completely by arguments already present in the supplied research corpus, which are independently reconstructed here. The other three remain unresolved; the accompanying theorems are partial results or consequences of additional hypotheses.

3.1 Question 1.6: the least square-clock constant

For an integer $c \geq 2$, let

$$\xi_c = \min\{A > 1 : \lfloor A^{c^n} \rfloor \text{ is prime for every } n \geq 1\}.$$

Existence is known. Saito's Question 1.6 in [1] asks whether ξ_2 is rational or irrational. The question is not answered here. The integer-tail mechanism does, however, give a complete result at the critical growth rate 2^n. This auxiliary result is proved directly; no claim of historical priority is made.

Theorem 3.1 (Irrationality of the reciprocal square-clock series)**.** *For every real number $A > 1$, the convergent series*

$$S(A) = \sum_{n=1}^{\infty} \frac{1}{\lfloor A^{2^n} \rfloor}$$

is irrational.

Proof. Write $a_n = \lfloor A^{2^n} \rfloor$ and $P_n = \prod_{j=1}^{n} a_j$. The initial values may equal 1; all values are positive and $a_n \to \infty$. Since

$$0 \leq 1 - \frac{a_n}{A^{2^n}} < A^{-2^n},$$

the infinite product $\Omega = \prod_{n\geq 1} a_n/A^{2^n}$ converges to a strictly positive number. Moreover, $a_{n+1} \geq a_n^2$. The first term dominates each tail:

$$\sum_{j>n} \frac{1}{a_j} = \frac{1}{a_{n+1}}(1 + o(1)),$$

because the remaining terms are at most $\sum_{\ell\geq 1} a_{n+1}^{-2^\ell} = O(a_{n+1}^{-2})$. As $\sum_{j=1}^{n} 2^j = 2^{n+1} - 2$, it follows that

$$P_n \sum_{j>n} \frac{1}{a_j} \longrightarrow A^{-2}\Omega > 0. \tag{3.1}$$

Suppose $S(A) = u/v$, where u, v are positive integers. Then

$$I_n = vP_n \sum_{j>n} \frac{1}{a_j} = uP_n - v\sum_{j=1}^{n} \frac{P_n}{a_j}$$

is a positive integer. By (3.1), this integer sequence converges, and hence is eventually an integer $m > 0$. The exact identity $I_{n+1} = a_{n+1}I_n - vP_n$ gives $m(a_{n+1} - 1) = vP_n$. Comparing successive indices yields

$$a_{n+2} - 1 = a_{n+1}(a_{n+1} - 1).$$

Thus $a_{n+2} = a_{n+1}^2 - a_{n+1} + 1 < a_{n+1}^2$ eventually, contradicting $a_{n+2} \geq a_{n+1}^2$. □

The theorem applies to $A = \xi_2$, but also to $A = 2$. Consequently its conclusion cannot determine the arithmetic nature of A. A proof of Question 1.6 still needs an implication from a rational hypothetical base to additional restrictions on its prime-floor sequence. The irrationality of $S(\xi_2)$ supplies no such implication by itself.

3.2 Question 1.7: the cubic Pisot alternative for Mills' constant

Write $M = \xi_3$. Saito's Question 1.7 in [1] asks whether M^{3^m} can be a cubic Pisot number for some positive integer m. A negative answer would prove M transcendental. The later reduction in [2, Theorem 1.7] shows that, if M is algebraic, some such power $\beta = M^{3^m}$ has three real conjugates

$$\beta,\ -\rho,\ \tau, \qquad 0 < \sigma := |\tau| < \rho < 1, \qquad \rho \le \min(\sigma^{17/23}, \beta^{-17/40}). \tag{3.2}$$

This is an external theorem. The conditional gap theorem below is a corpus result from [7], included with its proof because it specifies what an unconditional transcendence argument must exclude.

Theorem 3.2 (Arithmetic of a hypothetical algebraic Mills gap)**.** *Assume M algebraic and choose β as in (3.2). Set*

$$p_n = \lfloor \beta^{3^n} \rfloor, \qquad g_n = p_{n+1} - p_n^3, \qquad N = |\operatorname{Norm}(\beta)|.$$

On a sufficiently late tail,

$$g_n \mid g_{n+1}, \qquad \frac{9g_{n+1}}{g_n^3} \longrightarrow 1, \qquad \frac{g_n}{p_n} \longrightarrow \infty.$$

There is a transcendental real number $\kappa = (2 + \log\rho/\log\beta)/3$ such that

$$g_n \sim 3(p_n^3)^\kappa, \qquad \frac{1}{2} + \frac{\log N}{6\log\beta} < \kappa < \frac{21}{40}. \tag{3.3}$$

Proof. The sum of the two small conjugate powers is $-\rho^{3^n} + \tau^{3^n} \in (-1, 0)$ eventually. The integer trace is therefore $p_n = \operatorname{Tr}(\beta^{3^n})$. At a fixed index write X, Y, Z for the three conjugate powers and put $p = X + Y + Z$, $b = XY + XZ + YZ$, $c = XYZ$. These are integers. Cubing the roots gives

$$p' = p^3 - 3bp + 3c, \quad b' = b^3 - 3cbp + 3c^2, \quad c' = c^3.$$

In particular, $h_n = g_n/3 = c - bp$ and

$$h_n = -(X + Y)(X + Z)(Y + Z), \qquad h_{n+1} = h_n \prod_{\{U,V\}\subset\{X,Y,Z\}} (U^2 - UV + V^2).$$

The multiplier is a symmetric polynomial with integer coefficients in the algebraic integers X, Y, Z, so is an integer. It is positive because the three numbers are nonzero and real. This proves divisibility.

The inequalities between the conjugate moduli give

$$p_n \sim \beta^{3^n}, \qquad h_n \sim (\beta^2\rho)^{3^n}.$$

They imply the gap asymptotic and $9g_{n+1}/g_n^3 \to 1$. Also $1 \le N = \beta\rho\sigma < \beta\rho^2$. Thus $\rho > \sqrt{N/\beta}$, which gives the strict lower bound for κ; it also gives $\beta\rho > 1$, proving $g_n/p_n \to \infty$. The last inequality in (3.2) gives $\kappa \le 21/40$.

Let $t = \log\rho/\log\beta$. If t were rational, its negativity would imply $\beta^u\rho^v = 1$ for positive integers u, v. A transitive cubic Galois group contains a three-cycle. Choose its orientation to send $\beta \mapsto -\rho \mapsto \tau$. Applying it to the last identity and taking absolute values gives $\rho^u\sigma^v = 1$, impossible. Hence t is irrational. If t were algebraic, the Gelfond–Schneider theorem would make the positive real value $\beta^t = \rho$ transcendental [4]. Thus t, and therefore κ, is transcendental. The rational upper endpoint $21/40$ cannot occur. □

Corollary 3.3 (An exact conditional route to transcendence)**.** *Let $q_n = \lfloor M^{3^n} \rfloor$ be the actual Mills primes. If the limit*

$$\lim_{n\to\infty} \frac{\log(q_{n+1} - q_n^3)}{\log(q_n^3)}$$

exists and is an algebraic real number, then M is transcendental.

Proof. Under algebraicity, deleting finitely many indices identifies the limit with the transcendental κ in Theorem 3.2, a contradiction. □

No argument here proves that the limit in the corollary is algebraic or excludes the alternative in (3.2). The unconditional Mills question remains open in this paper. The implication RH $\Rightarrow M$ transcendental, and the stronger implication from the Density Hypothesis, are already Saito's Theorem 1.8 in [2]; they are not a new resolution.

3.3 Problem 1.1: composite values on the pure cubic clock

Problem 1.1 of [3] concerns every cubic Pisot number β, including a nonreal conjugate pair: must $\lfloor \beta^{3^n} \rfloor$ be composite infinitely often? Here cubic clock refers to the exponent 3^n, not to a pure cubic number field. We establish a substantial conditional family but do not answer the universal question. This simultaneous-window argument is an audited corpus result [8].

Lemma 3.4 (A three-adic trace limit)**.** *Let $f(X) = X^3 - uX^2 + vX - N$ be a monic integral cubic. Let T_n be the 3^n-th power sum of its roots, let B_n be the second elementary symmetric function of those powered roots, and put $E_n = N^{3^n}$. There is $\lambda \in \mathbb{Z}_3$ such that*

$$\lambda - T_n \in 3^{n+1}\mathbb{Z}_3 \qquad (n \geq 0).$$

The limit depends only on $(u, v, N) \pmod 3$.

Proof. Newton's identities give the integral polynomial iteration $(T_{n+1}, B_{n+1}, E_{n+1}) = F(T_n, B_n, E_n)$, where

$$F(x, y, z) = (x^3 - 3xy + 3z,\ y^3 - 3zxy + 3z^2,\ z^3).$$

It is the identity on $\mathbb{F}_3^3$. If two integer vectors agree modulo 3^r, $r \geq 1$, their images agree modulo 3^{r+1}. Indeed, every first derivative has coefficients divisible by 3, and terms of order at least two have the required divisibility. Successive iterates differ by a vector in $3^{n+1}\mathbb{Z}^3$. Completeness gives the limit and its precision. The same contraction applied to two initial states congruent modulo 3 proves the final assertion. □

Theorem 3.5 (A common progression of composite trace shifts)**.** *Suppose $T_n \to +\infty$ in the real topology. Let $\mathcal{H} \subset \mathbb{Z}$ be finite. If $\lambda + h \notin \{-1, 1\}$ for every $h \in \mathcal{H}$, then some arithmetic progression of indices contains only composite values $T_n + h$, simultaneously for all $h \in \mathcal{H}$, after finitely many initial exceptions.*

Proof. Choose a common large $n_0 \geq 1$ for which $Q_h = T_{n_0} + h > 1$ for every shift. When $3 \nmid \lambda + h$, also require

$$v_3(Q_h - 1) < n_0, \qquad v_3(Q_h + 1) < n_0.$$

This is possible because $\lambda + h - 1$ and $\lambda + h + 1$ are nonzero, and Lemma 3.4 eventually fixes their valuations. If $3 \mid \lambda + h$, then $3 \mid T_n + h$ at every index.

For each other shift, some prime factor ℓ_h of Q_h must satisfy

$$v_3(\ell_h^2 - 1) \leq n_0 - 1. \tag{3.4}$$

Otherwise every prime factor would be 1 or -1 modulo 3^{n_0}: for a prime different from 3, at most one of its predecessor and successor is divisible by 3. Their product would then make Q_h equal to 1 or -1 modulo 3^{n_0}, a contradiction. Here $\ell_h \neq 3$, since $3 \nmid Q_h$.

Every nonzero root of f over $\mathbb{F}_{\ell_h}$ belongs to an extension of degree at most three; its order divides

$$L_h = \operatorname{lcm}(\ell_h - 1, \ell_h^2 - 1, \ell_h^3 - 1).$$

For $\ell_h \equiv 1 \pmod 3$, the last factor has valuation $v_3(\ell_h - 1) + 1$; for $\ell_h \equiv -1 \pmod 3$, the only nonzero valuation among the three factors is $v_3(\ell_h + 1)$. Thus (3.4) implies $v_3(L_h) \leq n_0$. After raising the roots to 3^{n_0}, their nonzero orders are prime to 3. Choose a period $D_h \geq 1$ for cubing on these finitely many cyclic groups. Then

$$T_{n_0 + jD_h} \equiv T_{n_0} \pmod{\ell_h} \qquad (j \geq 0).$$

The power-sum argument also covers zero and repeated roots, with multiplicities. Take a common multiple D of all D_h, or $D = 1$ if no such primes were needed. At $n = n_0 + jD$, each trace shift has its fixed prime divisor. Real growth makes every divisor proper eventually. □

Corollary 3.6 (Pisot floor windows, including nonreal conjugates)**.** *Let β be any cubic Pisot number. If its trace limit λ is not an ordinary integer, then for every $H \geq 0$, all the numbers*

$$\lfloor \beta^{3^n} \rfloor + h, \qquad |h| \leq H,$$

are simultaneously composite on an eventual arithmetic progression of indices. Their common index set has positive lower natural density.

Proof. The two small conjugate powers have real sum tending to zero. The floor is therefore eventually T_n or $T_n - 1$, whether or not that sum changes sign. Apply Theorem 3.5 to every integer shift in $[-H - 1, H]$. Nonintegrality ensures $\lambda + h \neq \pm 1$ for all these shifts. □

For a concrete coefficient class, irreducibility of the cubic modulo 3 guarantees the required nonintegrality. In the unramified degree-three extension of $\mathbb{Q}_3$, the roots are units. A unit is its Teichmüller root of unity times a principal unit. The principal-unit part tends to 1 under 3^n-th powering. The three Teichmüller parts are permuted by Frobenius, so their sum is λ. Irreducibility forces their orders to be 13 or 26. For a primitive thirteenth root ζ,

$$\lambda = \pm(\zeta^a + \zeta^{3a} + \zeta^{9a}), \qquad a \not\equiv 0 \pmod{13}.$$

This cannot be rational: otherwise a polynomial of degree at most 12, with at most four nonzero coefficients, would vanish at ζ, but it cannot be a nonzero multiple of $1 + X + \cdots + X^{12}$. For example $X^3 - X - 1$ has no root modulo 3. Its real root $\beta > 1$ is Pisot, because the nonreal conjugate pair has modulus $\beta^{-1/2} < 1$.

The remaining obstruction is explicit. In a hypothetical algebraic Mills tower, the floor is eventually T_n. Its primality at every index forces $\lambda \in \{-1, 1\}$ by Theorem 3.5. These two possibilities are not excluded, so the partial result does not close Problem 1.1.

3.4 Problem 1.7: a non-reversible recurrence for every Pisot number

Problem 1.7 in [3] asks for a non-reversible inhomogeneous integer linear recurrence $R(n)$, chosen independently of the Pisot number α, for which $\lfloor \alpha^{R(n)} \rfloor$ is composite infinitely often for every α. The following answers the full formulation. It reconstructs [6], rather than presenting a discovery made in this paper. An elementary trace-period proof avoids any theorem on distribution of prime values.

Lemma 3.7 (Every residue occurs on a progression)**.** *Let $a \geq 2$, $b \geq 1$ be integers and $\varepsilon \in \{-1, 1\}$. For every $q \geq 1$ and $c \in \mathbb{Z}$, the solutions to*

$$R(n) := ba^n + \varepsilon n \equiv c \pmod{q}$$

include all sufficiently large indices in an arithmetic progression.

Proof. For $q > 1$, factor $q = uv$, where u contains exactly the prime-power factors supported on primes dividing a, and $\gcd(a, v) = 1$. The sequence $a^n \bmod q$ is eventually periodic with a period $t < q$: take $t = \mathrm{ord}_v(a) \leq \varphi(v) < v$ if $v > 1$, and $t = 1$ otherwise.

We prove arbitrarily late solutions exist by strong induction on q. The case $q = 1$ is immediate. Put $g = \gcd(t, q) < q$. By induction choose an arbitrarily large n, beyond the preperiod, with $R(n) \equiv c \pmod{g}$. Since $\gcd(\varepsilon t, q) = g$, there is a nonnegative j with $\varepsilon jt \equiv c - R(n) \pmod{q}$. Periodicity gives $R(n + jt) \equiv c \pmod{q}$. Finally $R(n) \bmod q$ is eventually periodic with period $\mathrm{lcm}(q, t)$, so any sufficiently late solution generates the claimed progression. □

Theorem 3.8 (Full solution of the non-reversible recurrence question)**.** *For every $a \geq 2, b \geq 1, \varepsilon \in \{-1, 1\}$, the sequence $R(n) = ba^n + \varepsilon n$, indexed by $n \geq 1$, is a positive, increasing, non-reversible inhomogeneous integer linear recurrence. No alternative integer recurrence representation makes it reversible. For every Pisot or Salem number α and every $H \geq 0$, all*

$$\lfloor \alpha^{R(n)} \rfloor + h, \qquad |h| \leq H,$$

are simultaneously composite on an eventual arithmetic progression of indices. In particular $R(n) = 3^n + n$ answers Saito's Problem 1.7.

Proof. The recurrence is

$$R(n + 2) = (a + 1)R(n + 1) - aR(n) + \varepsilon(1 - a).$$

Positivity and strict increase follow from $ba^n - n \geq 2^n - n \geq 1$ and $R(n + 1) - R(n) = b(a - 1)a^n + \varepsilon \geq 1$. To exclude a hidden reversible representation, suppose $Q(E)R$ is constant, where E is the shift and Q a monic integer polynomial. The coefficient of a^n forces $Q(a) = 0$, and that of n forces $Q(1) = 0$. Hence $(X - a)(X - 1)$ divides Q in $\mathbb{Z}[X]$. Its constant coefficient is divisible by a, so the trailing recurrence coefficient $-Q(0)$ cannot equal 1 or -1.

Let d be the degree of α, and set $T_m = \mathrm{Tr}(\alpha^m)$. This is an integer recurrence sequence. All conjugates except α have modulus at most 1, so

$$T_m \longrightarrow +\infty, \qquad |\lfloor \alpha^m \rfloor - T_m| \leq d.$$

The recurrence matrix has nonzero integer determinant, equal up to sign to the constant coefficient of the minimal polynomial. Modulo primes not dividing this determinant, the state is purely periodic. There are finitely many other primes, and each finite state space gives eventual periodicity. Thus an index s_0, independent of the prime, exists after which T_m is periodic modulo every prime, with a prime-dependent period.

Put $J = H + d$, and choose $m_0 \geq s_0$ so large that $T_{m_0} + j > 1$ for all integers $|j| \leq J$. For each j, choose a prime divisor ℓ_j of this integer. Let L be a common multiple of the finitely many trace periods modulo ℓ_j. For $m \geq m_0$ with $m \equiv m_0 \pmod{L}$,

$$\ell_j \mid T_m + j \qquad (|j| \leq J).$$

Real growth makes these divisors proper eventually. Lemma 3.7 supplies an eventual progression of n with $R(n) \equiv m_0 \pmod{L}$, and $R(n) \to \infty$. Finally each $\lfloor \alpha^{R(n)} \rfloor + h$, $|h| \leq H$, equals $T_{R(n)} + j$ for some integer $|j| \leq H + d$. The whole required window is composite on the same progression. □

The distinction from Problem 1.1 is exact: the added term εn makes every residue accessible and is used in the induction. The proof gives no such surjectivity for $R(n) = 3^n$. Dropping the added term drops an essential hypothesis.

3.5 Problem 1.8: every shifted dyadic Fibonacci sequence

Problem 1.8 of [3] asks whether $F_{2^r} + h$ is composite infinitely often for every fixed integer h, with the usual Fibonacci initial values $F_0 = 0, F_1 = 1$. The result below answers the complete question, with positive lower density. It independently reconstructs the existing corpus proof [5].

More generally, fix a positive odd integer P, and let

$$U_0 = 0, \quad U_1 = 1, \quad U_{n+2} = PU_{n+1} + U_n, \qquad D = P^2 + 4.$$

For $h \in \mathbb{Z}$, define the finite integer

$$B_{P,h} = \max\left\{v_2\Big(D(h-1)^2 + 3\Big), v_2\Big(D(h+1)^2 + 3\Big)\right\}. \tag{3.5}$$

Theorem 3.9 (Prime return and composite Fibonacci shifts)**.** *If $r \geq B_{P,h} + 1$ and $p = U_{2^r} + h$ is prime with $p \nmid 2D$, then some $k \geq 1$ satisfies*

$$p \mid U_{2^{r+jk}} + h \qquad (j \geq 1).$$

These later numbers are larger than p. For every positive odd P and every integer h, the set of $r \geq 1$ with $U_{2^r} + h$ composite has positive lower natural density. Taking $P = 1$ proves Saito's full Problem 1.8.

Proof. Let $V_0 = 2, V_1 = P, V_{n+2} = PV_{n+1} + V_n$, and put $f_r = U_{2^r}$, $\ell_r = V_{2^r}$, $r \geq 1$. For the roots α, β of $X^2 - PX - 1$,

$$U_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}, \qquad V_n = \alpha^n + \beta^n, \qquad \alpha\beta = -1.$$

Expansion at even indices gives

$$f_{r+1} = f_r\ell_r, \qquad \ell_{r+1} = \ell_r^2 - 2, \qquad \ell_r^2 - Df_r^2 = 4. \tag{3.6}$$

Here $f_1 = P$, $\ell_1 = P^2 + 2$, all f_r are odd, and all $\ell_r \equiv 3 \pmod 4$. Since $P^2 \equiv 1 \pmod 8$,

$$v_2(\ell_1 + 1) = 2, \qquad \ell_{r+1} + 1 = (\ell_r - 1)(\ell_r + 1),$$

and consequently $v_2(\ell_r + 1) = r + 1$. Also

$$f_{r+2} - f_r = f_r(\ell_r + 1)(\ell_r^2 - \ell_r - 1).$$

The first and last factors are odd; hence

$$v_2(f_{r+2} - f_r) = r + 1. \tag{3.7}$$

The two parity subsequences of f_r converge in $\mathbb{Z}_2$ to odd limits u_0, u_1. In the series for $u_{r \bmod 2} - f_r$, the first difference has valuation $r + 1$, and all later ones have valuation at least $r + 3$. Thus

$$v_2(f_r - u_{r \bmod 2}) = r + 1. \tag{3.8}$$

Since $\ell_r \to -1$, passing to the limit in (3.6) gives

$$u_1 = -u_0, \qquad Du_i^2 = -3.$$

Neither limit is an ordinary integer, because $D > 0$. For either limit u and every $c \in \mathbb{Z}$,

$$v_2(u - c) + v_2(u + c) = v_2(Dc^2 + 3),$$

by the product identity $(u - c)(u + c) = -(Dc^2 + 3)/D$. Both valuations are finite and nonnegative, so each is at most the right-hand side.

Now suppose $p = f_r + h$ is prime as in the theorem. Apply this bound at $c = 1 - h$ and $c = -1 - h$. By (3.8) and $r \geq B_{P,h} + 1$,

$$v_2(p - 1) \leq B_{P,h}, \qquad v_2(p + 1) \leq B_{P,h}. \tag{3.9}$$

Indeed, in $f_r - c = (f_r - u) + (u - c)$, the first term has strictly larger valuation.

Choose a root γ of $X^2 - PX - 1$ in $\mathbb{F}_{p^2}$, with multiplicative order t. The roots are nonzero and distinct since $p \nmid 2D$. If they lie in $\mathbb{F}_p$, then $t \mid p - 1$. Otherwise Frobenius interchanges the roots, whose product is -1, so $\gamma^p = -\gamma^{-1}$ and $t \mid 2(p + 1)$. In either case (3.9) implies $v_2(t) \leq B_{P,h} + 1 \leq r$. Thus $z = \gamma^{2^r}$ has odd order q. Choose $k \geq 1$ with $2^k \equiv 1 \pmod{q}$, taking $k = 1$ when $q = 1$. Then

$$\gamma^{2^{r+jk}} = \gamma^{2^r} \qquad (j \geq 0).$$

The other root is $\delta = -\gamma^{-1}$; because the exponents are even, the same identity holds for δ. Binet's formula is valid in $\mathbb{F}_{p^2}$, so

$$U_{2^{r+jk}} \equiv U_{2^r} \pmod{p}.$$

This proves the divisibility. Strict real increase of U_{2^r} makes the divisor proper.

Choose R so large that for $r \geq R$, the values $f_r + h$ exceed 1, exceed every prime divisor of $2D$, and satisfy $r \geq B_{P,h} + 1$. If no such value is prime, all are composite and the density is 1. Otherwise choose one prime value; the return just proved gives a composite progression of indices with density $1/k > 0$. No existence assumption for a prime seed is required. □

The late-index condition matters. For $P = 1, h = 4, r = 2$, one has $F_4 + 4 = 7$. Modulo 7, the dyadic state is $(f_2, \ell_2) = (3, 0)$. One iteration makes $f_r \equiv 0 \pmod{7}$ forever, so this early prime never divides a later shifted value. Here $B_{1,4} = 7$; the theorem does not apply. In contrast $h = -16$ gives $B_{1,-16} = 3$ and $F_{16} - 16 = 971$, a valid seed at $r = 4$. A root of $X^2 - X - 1$ modulo 971 is 174, of order 485, and $\mathrm{ord}_{485}(2) = 48$. Therefore

$$971 \mid F_{2^{4+48j}} - 16 \qquad (j \geq 1).$$

These exact examples illustrate the mechanism; the full all-shift theorem rests on the proof, not on finite testing.

3.6 What the five targets establish

Problems 1.7 and 1.8 of [3] are answered with their full original quantifiers by the two corpus proofs reconstructed above. Their status here is complete proof previously present in the supplied corpus, not a new discovery in this paper. The square-clock irrationality question, the Mills cubic-Pisot alternative, and the universal pure cubic-clock composite question are not closed. Their partial theorems identify an irrational derived series, a necessary transcendental gap exponent, and a three-adic exceptional set. None replaces the original question.

4 The external units conjecture for values of E-functions

The external problem. Fischler and Rivoal formulate the following problem as Conjecture 1.2 of [9]: if $\mathcal{E}$ is the ring of values of E-functions at algebraic points, is

$$\mathcal{E}^\times = \{\alpha e^\beta : \alpha \in \overline{\mathbb{Q}}^\times,\ \beta \in \overline{\mathbb{Q}}\}?$$

Their Theorem 1.3 proves the assertion for elements which are finite algebraic-coefficient exponential sums. We prove the conjecture on a larger explicit class: all E-values lying in the field generated by those exponential values and one exponential-integral value. The unrestricted conjecture is not concluded from this result.

Unless otherwise specified, E-functions have the strict arithmetic growth bounds C^{n+1} on coefficient conjugates and common denominators, as in the main formulation of [9]. The arguments also apply to the larger ring obtained from Siegel's original bounds $(n!)^\varepsilon$ for each $\varepsilon > 0$; the additional analytic input for that extension is specified below.

Put $A = \overline{\mathbb{Q}}$ and define

$$\mathcal{B} = A[e^\beta : \beta \in A], \qquad \mathcal{L} = \mathrm{Frac}(\mathcal{B}), \qquad \mathcal{K} = \mathrm{Frac}(\mathcal{E}).$$

The ring $\mathcal{B}$ already contains every $e^{-\beta}$. For $a \in A \setminus \{0\}$ write

$$\eta_a = \mathrm{Ein}(a), \qquad \mathrm{Ein}(z) = \sum_{n\geq 1} \frac{(-1)^{n-1} z^n}{n\, n!}.$$

No arithmetic assumption on Euler's constant is made in this section.

4.1 A polynomial parameter for repeated winding

We recall the analytic inputs and state exactly where they enter. Choose a direction in $(-\pi/2, \pi/2)$ avoiding all anti-Stokes directions of E-functions; this is possible because the family of E-functions is countable and each has finitely many such directions. For this fixed direction, the canonical decomposition of an E-function has the form

$$f(z) = \sum_\rho f_\rho(z) e^{\rho z}, \qquad \rho \in A,$$

with finite support. The components $e^{\rho z} f_\rho(z)$ solve an E-operator and continue along all paths in $\mathbb{C}^\times$ [9, Theorem 2.3 and the paragraph preceding it]. This is a homogeneous annihilating operator for the full components, even if a smaller inhomogeneous equation has also been used to describe the original function. Such an operator is regular singular at zero with rational exponents [10, §3.2, proof of Proposition 3, p. 41]. Consequently its local monodromy is quasi-unipotent. Multiplication by the single-valued, nowhere-zero factor $e^{-\rho z}$ identifies the monodromy on $e^{\rho z} f_\rho$ with that on f_ρ.

For the broader Siegel definition, the same existence and local structure follow from [12, Proposition 16 and Theorem 7(a,b,d)]; the continuation, canonical decomposition, and zero-division inputs extend as explained in [9, pp. 3–4]. In particular, the quasi-unipotence used below does not require the unproved equivalence of the strict and broad definitions.

Let M^j denote continuation around a based loop at 1 of winding number j about zero. Define

$$\Phi_j(f(1))(X) = \sum_\rho (M^j f_\rho)(1) e^{\rho X}.$$

These are injective A-algebra homomorphisms from $\mathcal{E}$ to

$$\mathcal{R} = \left\{ \sum_{\rho \in A} c_\rho e^{\rho X} : c_\rho \in \mathbb{C},\ \text{finite support} \right\},$$

and satisfy $\Phi_j(\xi)(1) = \xi$ and $\Phi_j(e^\beta) = e^{\beta X}$. Here is the well-definedness argument, including its continuation step. If $f(1) = g(1)$, Beukers' theorem [11] gives an E-function h with $f - g = (z-1)h$. Canonical uniqueness gives $f_\rho - g_\rho = (z-1)h_\rho$. After continuation, h_ρ remains holomorphic at 1, so the corresponding continued component values agree. Componentwise addition and convolution for products give the homomorphism property. Continuing the sum recovers the entire, single-valued function f, so evaluation at $X = 1$ is a left inverse.

Lemma 4.1 (Polynomial winding on a finite algebra)**.** *For every finitely generated A-subalgebra $S \subset \mathcal{E}$, there are an integer $m \geq 1$ and an injective A-algebra homomorphism*

$$\Psi : S \longrightarrow \mathcal{R}[T]$$

such that $\Psi(\xi)|_{T=j} = \Phi_{mj}(\xi)$ for every integer j. It extends to the fraction field of S.

Proof. Choose representing E-functions for finitely many generators of S and their finitely many canonical components. A common positive integer m makes every relevant monodromy operator M^m unipotent. If $M^m = I + N$ with N nilpotent, then

$$(M^m)^j = \sum_{k \geq 0} \binom{j}{k} N^k$$

is a finite sum and is polynomial in j, for all integers j. Every coefficient of $\Phi_{mj}(\xi)$ for a chosen generator is therefore polynomial in j. Replace j by an indeterminate T.

If a polynomial relation with coefficients in A holds among the generators, its images vanish after setting T equal to every integer. A polynomial over the domain $\mathcal{R}$ with infinitely many roots is zero. Hence the assignment is well-defined and multiplicative on S. Evaluation at $X = 1$ gives the constant polynomial ξ because this holds at every integer T. Thus the map is injective and extends to fractions. No common monodromy exponent for all of $\mathcal{E}$ is asserted or needed. □

For fixed nonzero algebraic a, the canonical components of $\mathrm{Ein}(az)$, with a compatible branch, have the form

$$f_0(z) = C + \log z, \qquad f_{-a}(z) = e^{az}(\mathrm{Ein}(az) - C - \log z).$$

The constant C depends on the chosen branch. The two phases are 0 and $-a$. Continuing adds $2\pi i j$ to f_0 and subtracts $2\pi i j e^{az}$ from f_{-a}. Writing

$$c = e^a, \qquad B = c(\eta_a - C), \qquad \lambda = 2\pi i m,$$

the polynomial map of Lemma 4.1, on an algebra containing c and η_a, consequently satisfies

$$\Psi(c) = e^{aX}, \qquad \Psi(\eta_a) = C + Be^{-aX} + \lambda T(1 - ce^{-aX}). \tag{4.1}$$

The symbols C, B, c on the right are complex coefficients. Their use does not assert that they are additional elements of the domain.

Lemma 4.2 (Finite exponential phases with a parameter)**.** *Let $a_1, \dots, a_r \in A$ be linearly independent over $\mathbb{Q}$, and put $U_i = e^{a_i X}$. If $W \in \mathrm{Frac}(\mathcal{R}[T])$ is algebraic over $\mathbb{C}(T, U_1, \dots, U_r)$, then for some $q \geq 1$,*

$$W \in \mathbb{C}(T, e^{a_1 X/q}, \dots, e^{a_r X/q}).$$

Moreover,

$$\mathcal{R}[T] \cap \mathbb{C}(T, U_1, \dots, U_r) = \mathbb{C}[T, U_1^{\pm 1}, \dots, U_r^{\pm 1}]. \tag{4.2}$$

Proof. Extend $a_1, \dots, a_r$ to a rational basis for the span of the finitely many phases occurring in a numerator and denominator for W, and clear denominators. The resulting exponential variables are algebraically independent over $\mathbb{C}(T)$, by linear independence of exponential functions with distinct phases. The additional basis variables give a purely transcendental extension and cannot contribute to an element algebraic over the indicated smaller field. This proves the first assertion.

In the ring case the expression initially is a Laurent polynomial in these variables with coefficients in $\mathbb{C}[T]$. Independence removes all monomials involving an additional basis variable. Invariance under the finite Kummer transformations of $e^{a_i X/q}$ then forces each surviving exponent to be divisible by q. This proves (4.2). □

We also use the elementary constant-extension fact that an irreducible polynomial over $A(U_1, \dots, U_r, V)$ remains irreducible over $\mathbb{C}(U_1, \dots, U_r, V)$. After clearing denominators this is absolute irreducibility over the algebraically closed field A. It follows, for example, by expressing a hypothetical factorization as a finite polynomial system and applying the weak Nullstellensatz. In particular, adjoining complex constants preserves the degree of a finite extension of such a rational function field.

4.2 Complete unit classification in an explicit enlarged field

Theorem 4.3 (Exponential values with one Ein value)**.** *Let $a \in A \setminus \{0\}$ and put $F_a = \mathcal{L}(\eta_a)$. Then:*

1. *η_a is transcendental over $\mathcal{L}$.*
2. *F_a is relatively algebraically closed in $\mathcal{K}$.*
3. *$\mathcal{E} \cap F_a = \mathcal{B}[\eta_a]$.*
4. $$\mathcal{E}^\times \cap F_a = \{\alpha e^\beta : \alpha \in A^\times,\ \beta \in A\}.$$

Thus the external units conjecture holds for every E-value that is algebraic over F_a, as well as for every E-value in F_a itself.

Proof. Every expression under consideration uses only finitely many exponential values. Include a among their phases, choose a rational basis beginning with a, and divide all its members by a common positive integer d that clears the coordinates of the finitely many phases. The resulting basis $b_1, \dots, b_r$ satisfies $a = db_1$ and every phase under consideration is an integer combination of the b_i. Adjoin the corresponding values e^{b_i} and their inverses. Apply Lemma 4.1 to an algebra containing the required E-values. Put $U_i = e^{b_i X}$ and write $V = \Psi(\eta_a)$. Equation (4.1) becomes

$$V = C + B/U_1^d + \lambda T(1 - c/U_1^d), \qquad c = e^a. \tag{4.3}$$

Its coefficient of T is nonzero. Hence $U_1, \dots, U_r, V$ are algebraically independent over $\mathbb{C}$ and

$$\mathbb{C}(U_1, \dots, U_r, V) = \mathbb{C}(U_1, \dots, U_r, T).$$

A polynomial relation over A among the actual values $e^{b_1}, \dots, e^{b_r}, \eta_a$ would give one among these independent images. This proves the first assertion, including transcendence over the union of all the finite exponential fields.

For the second assertion, let $v \in \mathcal{K}$ be algebraic over F_a. Its algebraic equation uses finitely many exponential values, so enlarge the finite basis as above. Include a numerator and a denominator representing v among the generators used to construct Ψ. Its image W is algebraic over $\mathbb{C}(T, U_1, \dots, U_r)$. Lemma 4.2 gives

$$W \in \mathbb{C}(T, e^{b_1 X/q}, \dots, e^{b_r X/q})$$

for some $q \geq 1$. Adjoin the actual exponential roots $e^{b_i/q}$; their monodromy is trivial, so the same construction extends to them. Set

$$F_q = A(e^{b_1/q}, \dots, e^{b_r/q}, \eta_a).$$

The images of its $r+1$ generators are algebraically independent over $\mathbb{C}$; solving (4.3) for T shows that $W \in \mathbb{C}\Psi(F_q)$. The degree of v over F_q is preserved under extension of constants from A to $\mathbb{C}$. That degree is therefore one, and $v \in F_q \subset F_a$.

For the ring intersection, let $x \in \mathcal{E} \cap F_a$, and choose a finite basis as before so that

$$x = R(e^{b_1}, \dots, e^{b_r}, \eta_a), \qquad R \in A(U_1, \dots, U_r, V).$$

Write $R = P/Q$ with coprime P, Q in the Laurent polynomial ring

$$A[U_1^{\pm 1}, \dots, U_r^{\pm 1}, V].$$

They remain coprime after extending constants to $\mathbb{C}$. Let σ be the substitution in (4.3). By (4.2),

$$\sigma(P)/\sigma(Q) = \Psi(x) \in \mathbb{C}[T, U_1^{\pm 1}, \dots, U_r^{\pm 1}].$$

After inverting $U_1^d - c$, the substitution σ is an isomorphism of the two Laurent polynomial rings, because

$$T = \frac{U_1^d(V - C) - B}{\lambda(U_1^d - c)}.$$

Consequently every common irreducible factor of $\sigma(P)$ and $\sigma(Q)$ would have to divide $U_1^d - c$.

For each root ζ of $U_1^d - c$, at $U_1 = \zeta$ the substituted value of V is $C + B/c = \eta_a$, independently of T. A nonzero Laurent polynomial over A cannot vanish identically under $(U_1, V) = (\zeta, \eta_a)$: the numbers ζ and η_a are algebraically independent over A, since $\zeta^d = c$ and the first part of the proof makes c, η_a algebraically independent. Thus no factor $U_1 - \zeta$ divides $\sigma(Q)$. The substituted numerator and denominator are coprime. Since their quotient lies in the target ring, $\sigma(Q)$ must be a unit there, of the form $d_0 U_1^{n_1} \cdots U_r^{n_r}$ with $d_0 \in \mathbb{C}^\times$. Injectivity of σ then gives the same identity for Q, and its algebraic coefficients imply $d_0 \in A^\times$. The original denominator is a unit. Therefore $x \in A[e^{\pm b_1}, \dots, e^{\pm b_r}, \eta_a] \subset \mathcal{B}[\eta_a]$. The reverse inclusion is immediate.

Finally, if $x \in \mathcal{E}^\times \cap F_a$, both x and x^{-1} lie in $\mathcal{B}[\eta_a]$. Because η_a is transcendental over $\mathcal{L}$, the units of this polynomial ring are exactly the units of $\mathcal{B}$. A unit and its inverse in $\mathcal{B}$ involve only finitely many phases. Clearing their rational coordinates identifies them with mutually inverse Laurent polynomials in algebraically independent exponential values. Such units are precisely monomials, so $x = \alpha e^\beta$ with $\alpha \in A^\times$ and $\beta \in A$. Their inverses plainly belong to $\mathcal{E}$. The final sentence of the theorem follows from relative algebraic closure. □

Corollary 4.4 (Explicit non-units and non-members)**.** *Let $P \in \mathcal{B}[Y]$ be nonconstant. Then $P(\eta_a)$ is a nonzero non-unit of $\mathcal{E}$, and*

$$\frac{1}{P(\eta_a)} \notin \mathcal{E}.$$

More generally, a rational function in η_a with coefficients in $\mathcal{L}$ is an E-value exactly when it belongs to $\mathcal{B}[\eta_a]$.

Proof. The first assertion follows from the transcendence of η_a over $\mathcal{L}$ and the unit classification. The second is the proved ring intersection. □

Example 4.5. Writing $\eta = \mathrm{Ein}(1)$, the numbers

$$\eta^2 + e\eta + e^{\sqrt{2}}, \qquad e^{\sqrt{3}}\eta + e^{-\sqrt{2}}, \qquad \eta - e^a \quad (a \in A)$$

are nonzero E-values whose reciprocals are not E-values. These examples use the same complete classification; they are not additional external conjectures counted as separately solved.

4.3 Several independent Ein scales

The complementary family below allows several exponential-integral values without adjoining all exponential values to their base field. This distinction is essential: the two field constructions are not silently combined.

Theorem 4.6 (Several Ein values)**.** *Let $a_1, \dots, a_r \in A \setminus \{0\}$ be linearly independent over $\mathbb{Q}$, with no reality or positivity restriction. Put*

$$F = A(\eta_{a_1}, \dots, \eta_{a_r}), \qquad D = A[\eta_{a_1}, \dots, \eta_{a_r}].$$

The displayed generators are algebraically independent, F is relatively algebraically closed in $\mathcal{K}$, and

$$\mathcal{E} \cap F = D, \qquad \mathcal{E}^\times \cap F = A^\times.$$

Proof. For a fixed common lateral direction, write

$$\Phi_j(\eta_{a_i}) = C_i + j\tau + (B_i - j\tau e^{a_i})e^{-a_i X}, \qquad \tau = 2\pi i.$$

There is at most one integer j making any specified exponential coefficient zero. Choose two distinct integers j_0, j_1 avoiding all these finitely many exceptions. In either image the r generators are affine functions of r algebraically independent exponential variables. This proves their algebraic independence over A.

Let $v \in \mathcal{K}$ be algebraic over F, and identify the generators abstractly with $U_1, \dots, U_r$. In either image, finite-phase descent places the resulting extension, after extending constants to $\mathbb{C}$, inside a translated Kummer extension

$$\mathbb{C}(Z_1, \dots, Z_r)/\mathbb{C}(U_1, \dots, U_r), \qquad U_i = C_i + j\tau + (B_i - j\tau e^{a_i})Z_i^q.$$

Its group is $(\mathbb{Z}/q\mathbb{Z})^r$. The inertia at the generic point of $U_i = C_i + j\tau$ is its ith coordinate subgroup, and these inertia groups generate the entire group. Every nontrivial intermediate extension is therefore ramified along at least one of these finite hyperplanes.

The two realizations describe the same abstract extension of $\mathbb{C}(U_1, \dots, U_r)$, because extension of constants preserves its degree and defining polynomial. The second Kummer realization is unramified at the generic point of every finite hyperplane from the first: for the same coordinate the constants differ by $(j_1 - j_0)\tau$, and hyperplanes in different coordinates are distinct. Thus all coordinate inertia groups of the first realization act trivially on the intermediate field. That field is the base field, so the original degree is one and $v \in F$.

For the ring assertion, write an element of $F \cap \mathcal{E}$ as $P(\eta_{a_1}, \dots, \eta_{a_r})/Q(\eta_{a_1}, \dots, \eta_{a_r})$ with coprime $P, Q \in A[U_1, \dots, U_r]$. Finite-phase descent in the ring case implies that its image is

a Laurent polynomial in the variables $e^{-a_i X}$. Its denominator can therefore have irreducible factors only among $U_i - (C_i + j\tau)$. Applying this conclusion for both selected windings leaves no possible factor: the two lists of hyperplanes are disjoint. Hence Q is constant and the intersection is D. If an element of F is a unit of $\mathcal{E}$, it and its reciprocal belong to the polynomial ring D. Its only units are $A^\times$. □

What remains in the external problem. Theorems 4.3 and 4.6 completely classify units in their specified classes. An arbitrary E-value need not be algebraic over any of the fields used here. No reduction of every unit of $\mathcal{E}$ to those fields has been proved. That missing global step is precisely why Conjecture 1.2 is recorded as a partial solution in the ten-problem register.

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The external question is whether

$$S(f) = \sum_{n\geq 1} \frac{n!}{(n + f(n))!} \tag{5.1}$$

is irrational whenever $f : \mathbb{N} \to \mathbb{N}$ is nondecreasing and tends to infinity. Crmarić and Kovač [14] disprove the version without monotonicity: its values fill $(0, \infty)$. They explicitly retain the monotone question and prove that its set of possible values has measure zero. We do not close the monotone question. We prove a concrete subfamily, with a bounded-numerator strengthening.

A necessary historical correction. The 2025 paper also singled out $f(n) = n$. That example must not be counted as a new solution here. Lambrinoudis's primary repository [15] credits Crmarić–Kovač's elementary proof, posted by Kovač in July 2026, and explains that its transcendence follows from an earlier Siegel–Shidlovskii argument. The same repository contains broader affine-family claims. The results below use neither those claims nor their analytic inputs.

Lemma 5.1 (Clearing denominators in a short window)**.** *Let $f : \mathbb{N} \to \mathbb{N}$ be nondecreasing, and write*

$$g_n = n + f(n), \qquad D_n = \frac{g_n!}{n!}, \qquad r_n = \frac{D_{n+1}}{D_n}.$$

Suppose that, for all sufficiently large N, an integer $h_N \in \{1, \dots, N-1\}$ satisfies

$$g_N - g_{N-h_N} \geq N. \tag{5.2}$$

Then the integer

$$L_N = \frac{g_N!}{(N-h_N)!} = D_N H_N, \qquad H_N = \frac{N!}{(N-h_N)!},$$

is divisible by every D_k with $1 \leq k \leq N$. If, in addition,

$$\frac{H_N}{r_N} \longrightarrow 0, \tag{5.3}$$

then, for every bounded integer sequence (u_n) having infinitely many nonzero terms, the convergent series

$$\sum_{n\geq 1} \frac{u_n}{D_n}$$

is irrational.

Proof. If $k \le N - h_N$, then $g_N - g_k \ge N$. The numerator $g_N!/g_k!$ contains a product of N consecutive positive integers, and such a product is divisible by $N!$. Consequently

$$\frac{D_N}{D_k} = \frac{g_N!}{g_k!}\frac{k!}{N!} \in \mathbb{Z}.$$

If $N - h_N < k \le N$, then instead

$$\frac{L_N}{D_k} = \frac{g_N!}{g_k!}\frac{k!}{(N-h_N)!} \in \mathbb{Z}.$$

This proves the divisibility assertion.

Since $H_N \ge 1$, condition (5.3) implies $r_N \to \infty$. Hence all the series in question converge absolutely. Put $B = \sup_n |u_n|$, which is a positive integer, and choose an arbitrarily large N with $u_{N+1} \ne 0$. Since the successive ratios tend to infinity, for all sufficiently large such N the tail satisfies

$$\frac{1}{2D_{N+1}} \le \left|\sum_{k>N} \frac{u_k}{D_k}\right| \le \frac{2B}{D_{N+1}}.$$

For the lower bound, the first nonzero term has magnitude at least $1/D_{N+1}$, while the sum of subsequent magnitudes is $o(1/D_{N+1})$. The upper bound follows from a geometric majorant.

If the whole sum were $a/b \in \mathbb{Q}$, with $b > 0$, then

$$bL_N\left(\frac{a}{b} - \sum_{k\le N}\frac{u_k}{D_k}\right)$$

would be a nonzero integer whose absolute value is at most $2bBH_N/r_N \to 0$. This is impossible. □

Theorem 5.2 (A power-growth subfamily of the monotone question)**.** *Let $c > 0$, and let $f : \mathbb{N} \to \mathbb{N}$ agree eventually with*

$$f(n) = \lfloor cn^\alpha \rfloor.$$

The series (5.1) is irrational in each of the following ranges:

$$\alpha > \frac{3}{2}; \qquad \text{or} \qquad \alpha = \frac{3}{2}, \quad c > \left(\frac{2}{3}\right)^{3/2}. \tag{5.4}$$

More strongly, replacing the numerator $1/D_n$ by u_n/D_n for any bounded integer sequence with infinitely many nonzero terms preserves irrationality.

Proof. Changing finitely many terms changes the sum by a rational number, so we may begin where f is positive and nondecreasing. With the notation of Lemma 5.1,

$$g_n \sim cn^\alpha, \qquad g_{n+1} - g_n = (c\alpha + o(1))n^{\alpha-1}.$$

The exact quotient is

$$r_n = \frac{\prod_{j=g_n+1}^{g_{n+1}} j}{n+1}.$$

All factors in this numerator are asymptotic to cn^α; therefore

$$\log r_n = (c\alpha^2 + o(1))n^{\alpha-1}\log n. \tag{5.5}$$

First suppose $3/2 \le \alpha < 2$. Choose $t > 1/(c\alpha)$ and put $h_N = \lceil tN^{2-\alpha} \rceil$. Taylor's formula and the bounded floor error give

$$g_N - g_{N-h_N} = (c\alpha t + o(1))N > N$$

for large N, so (5.2) holds. Moreover

$$\log H_N \le h_N \log N = (t + o(1))N^{2-\alpha} \log N. \tag{5.6}$$

When $\alpha > 3/2$, comparison with (5.5) proves $H_N/r_N \to 0$. At $\alpha = 3/2$, the hypothesis on c is exactly the condition $1/(c\alpha) < c\alpha^2$. We can then choose t strictly between these two numbers, and the same comparison again proves the limit.

For $\alpha = 2$, take a fixed integer $h > 1/(2c)$. Then $g_N - g_{N-h} \sim 2chN > N$, whereas $\log H_N = O(\log N) = o(\log r_N)$. For $\alpha > 2$, take $h = 1$; the same conclusions follow directly. Lemma 5.1 completes all cases. □

Corollary 5.3 (Polynomial factorial shifts)**.** *Let f be a polynomial of degree at least two with positive leading coefficient and integer values at integers. Begin the series where $f(n) > 0$. Then*

$$\sum_n \frac{n!}{(n + f(n))!}$$

is irrational, as is every version with bounded integer numerators having infinitely many nonzero terms.

Proof. If $f(n) = cn^d + O(n^{d-1})$, then $g_{n+1} - g_n = cdn^{d-1} + O(n^{d-2})$. For $d = 2$ choose a fixed integer $h > 1/(2c)$; for $d > 2$ choose $h = 1$. Condition (5.2) follows, while $\log H_N = O(\log N)$ and $\log r_N = (cd^2 + o(1))N^{d-1} \log N$. Apply Lemma 5.1. □

Corollary 5.4 (Disjoint subseries)**.** *Under either Theorem 5.2 or Corollary 5.3, let $A_1, \dots, A_q$ be pairwise disjoint infinite subsets of $\mathbb{N}$. Then the numbers*

$$1, \quad \sum_{n \in A_1} \frac{1}{D_n}, \quad \dots, \quad \sum_{n \in A_q} \frac{1}{D_n}$$

are linearly independent over $\mathbb{Q}$.

Proof. An integer linear combination of the subseries with at least one nonzero coefficient is a series with bounded integer numerators and infinitely many nonzero terms. The preceding results make that combination irrational, so it cannot equal a rational constant. □

Relation to the original tail method. For $f(n) = n$, one has $D_n = (2n)!/n!$ and $D_{n+1}/D_n = 4n + 2 \in \mathbb{Z}$. If its sum were rational, the positive integral multiple of its Nth tail would be bounded by $b/(4N + 1) \to 0$. This recovers the already known elementary proof. For general monotone f, those successive quotients need not be integers. Lemma 5.1 supplies a different common denominator and pays the explicit factor H_N. The remaining barrier is to make that factor smaller than r_N for all monotone $f \to \infty$; the present proof does not do so. The constants in (5.4) describe this argument's range and are not claimed to be optimal boundaries of irrationality.

6 The Meijer identity for $\gamma + \log z$

The ninth target is the analytic identity requested in [16, Section 9.2(1)]. That paragraph also asks about recurrence comparisons; the result here resolves its identity component, with all ranks and all arguments on the principal branch. It does not settle the broader dynamical conjecture treated in the next section.

For $p \geq 2$, put

$$G_p(z) = G^{p+1,p}_{p,p+1}\left(\begin{matrix}0,\dots,0\\0,\dots,0\end{matrix}\middle| z\right), \qquad \theta = z\frac{d}{dz}.$$

We work first with $z > 0$ and then continue to $\mathscr{U} = \mathbb{C} \setminus (-\infty, 0]$, using $\operatorname{Log} z$. The Barnes convention is the clockwise loop around nonnegative integer poles [13, Section 16.17]:

$$G_p(z) = \frac{1}{2\pi i}\int_L \Phi_p(s) z^s \, ds, \qquad \Phi_p(s) = \Gamma(-s)^{p+1}\Gamma(1+s)^p. \tag{6.1}$$

Theorem 6.1 (The full argument-dependent identity)**.** *Define the differential operators*

$$\mathcal{N}_p(z) = z(1+\theta)^{p-2} + (-1)^p\theta^{p-1},$$
$$\mathcal{D}_p(z) = z(1+\theta)^{p-1} + (-1)^p\theta^p.$$

In these expressions z means multiplication on the left. For every $z \in \mathscr{U}$,

$$\mathcal{N}_p(z)G_p(z) = \gamma + \operatorname{Log} z, \qquad \mathcal{D}_p(z)G_p(z) = 1. \tag{6.2}$$

At $z = 1$ these recover the operators of [16, Proposition 1].

First proof: differential equation and its boundary constants. The standard Meijer equation [13, Section 16.21] is

$$[z(1+\theta)^p + (-1)^p\theta^{p+1}]G_p = 0.$$

The operator identity $\theta z = z(\theta + 1)$ gives

$$\theta\mathcal{N}_p = \mathcal{D}_p, \qquad \theta\mathcal{D}_p = z(1+\theta)^p + (-1)^p\theta^{p+1}.$$

Therefore $\mathcal{D}_pG_p$ is constant, and $\mathcal{N}_pG_p$ is affine in $\operatorname{Log} z$.

To determine the constants, the expansion at the Barnes pole zero is

$$\Phi_p(s) = (-1)^{p+1}s^{-p-1}(1 + \gamma s + O(s^2)).$$

Clockwise orientation contributes the negative residue. As $z \to 0^+$ this yields

$$G_p(z) = (-1)^p\left\{\frac{(\log z)^p}{p!} + \gamma\frac{(\log z)^{p-1}}{(p-1)!} + P_{p-2}(\log z)\right\} + O\big(z(1+|\log z|^p)\big),$$

where P_{p-2} is a polynomial of degree at most $p-2$. The same remainder estimate holds after any fixed number of Euler derivatives. One may see this directly from the convergent residue expansion: its nth summand is z^n times a polynomial in $\log z$ of degree at most p, with coefficients bounded by $O((1+\log(n+1))^p/n!)$ up to fixed polynomial factors in n.

It follows that $\mathcal{D}_pG_p \to 1$ and $\mathcal{N}_pG_p - \log z \to \gamma$. These determine both constants. Analytic continuation gives the result on $\mathscr{U}$. □

6.1 A residue hierarchy beyond the first two identities

The same mechanism produces higher polynomials in $\gamma + \operatorname{Log} z$ and zeta values. This extension also supplies an independent telescoping verification of Theorem 6.1.

Lemma 6.2 (Weighted Barnes telescoping)**.** *For every polynomial $Q \in \mathbb{C}[s]$, define*

$$\mathcal{T}_Q = zQ(\theta) + (-1)^p \theta Q(\theta - 1).$$

Then

$$\mathcal{T}_Q G_p(z) = z \operatorname{Res}_{s=-1}\big(Q(s)\Phi_p(s)z^s\big). \tag{6.3}$$

Proof. The functional equation of Γ gives $\Phi_p(s-1) = (-1)^{p+1} s \Phi_p(s)$. If $F_z(s) = Q(s)\Phi_p(s)z^s$, the Barnes integrand for $\mathcal{T}_Q G_p$ is consequently $z(F_z(s) - F_z(s-1))$. Summing its negative residues at $s = 0, \dots, N$ telescopes to

$$z \operatorname{Res}_{s=-1} F_z(s) - z \operatorname{Res}_{s=N} F_z(s).$$

The last residue tends to zero locally uniformly on $\mathscr{U}$: reflection gives $\Phi_p(s) = (-\pi)^p \Gamma(-s) \csc^p(\pi s)$, so its coefficients at $s = N$ are bounded by a fixed power of $1 + \log(N+1)$ times $1/N!$. Multiplication by $Q(s)z^s$ adds only polynomial factors in N and the factor $|z|^N$ on each compact set. This also justifies termwise application of the finite differential operator, proving the identity. □

Theorem 6.3 (A zeta-polynomial hierarchy)**.** *For $0 \le j \le p-1$, set*

$$\mathcal{T}_{p,j} = z(1+\theta)^{p-1-j} + (-1)^p \theta^{p-j}, \qquad T = \gamma + \operatorname{Log} z.$$

Then

$$\mathcal{T}_{p,j} G_p(z) = [t^j] \exp\left(Tt + \sum_{m \ge 2} \frac{1 + p(1 + (-1)^m)}{m} \zeta(m) t^m \right). \tag{6.4}$$

The first four resulting expressions, when their indices are available, are

$$1, \quad T, \quad \frac{T^2 + (2p+1)\zeta(2)}{2}, \quad \frac{T^3 + 3(2p+1)\zeta(2)T + 2\zeta(3)}{6}.$$

Proof. Use Lemma 6.2 with $Q(s) = (s+1)^{p-1-j}$. At $s = -1 + t$,

$$\Phi_p(-1+t) = t^{-p}\Gamma(1-t)\left(\frac{\pi t}{\sin \pi t}\right)^p.$$

Thus the right side of (6.3) is

$$[t^j] e^{t \operatorname{Log} z} \Gamma(1-t) \left(\frac{\pi t}{\sin \pi t}\right)^p.$$

The convergent local expansions

$$\log \Gamma(1-t) = \gamma t + \sum_{m \ge 2} \frac{\zeta(m)}{m} t^m, \quad \log \frac{\pi t}{\sin \pi t} = \sum_{m \ge 2} \frac{1 + (-1)^m}{m} \zeta(m) t^m$$

give (6.4). Only finitely many coefficients are used for each j. □

Corollary 6.4 (Exact kernel of weighted telescoping)**.** *For fixed p, the map $Q \mapsto \mathcal{T}_Q G_p$ has image $\mathbb{C}[\gamma + \operatorname{Log} z]_{\le p-1}$ and kernel $((s+1)^p) \subset \mathbb{C}[s]$. More explicitly, if $H_j(T)$ denotes the coefficient polynomial in (6.4), then*

$$\mathcal{T}_Q G_p(z) = \sum_{\ell=0}^{p-1} \frac{Q^{(\ell)}(-1)}{\ell!} H_{p-1-\ell}(\gamma + \operatorname{Log} z).$$

Proof. Taylor-expand $Q(-1+t)$ in the residue formula. The residue uses only its terms of degrees at most $p-1$. The polynomials $H_0, \dots, H_{p-1}$ have distinct degrees and leading coefficients $1/j!$, so they form a basis of the stated image. Their linear independence identifies the kernel exactly. □

Remark 6.5 (Scope of the resolution). The $j = 0, 1$ cases are precisely the denominator and numerator identities, now for arbitrary z. The factor z on the left is essential. Substituting arbitrary z into the operators originally written only for evaluation at 1 would miss it. These are exact analytic identities. They do not establish the irrationality of γ, nor an integer-denominator estimate for any recurrence derived from them. No claim of historical priority for identities equivalent to known special-function formulas is made.

7 The all-trajectory CMF conjecture and an all-rank scalar theorem

The tenth target is Conjecture 1 of [16]. For rank $r \geq 3$, its trajectories are

$$\mathbf{t} = (a_1, \dots, a_{r-1}; b_1, \dots, b_r), \quad a_i \leq 0, \quad b_j \geq 1,$$

with the smallest b_j occurring at least twice. The conjecture asks for generic dual-state convergence along every such trajectory. The source proves the rank-three subfamily $b_1 = b_2 = b_3 = 1$. We do not prove its all-rank, all-trajectory assertion. We establish below a directly analyzable scalar approximation family and explain the precise extra step needed to connect it to that assertion.

7.1 A positive binomial family at every rank

For an integer $r \geq 2$, a fixed real $z > 0$, and $0 \leq m \leq n$, set

$$w_{n,m}(z) = \binom{n}{m}^r (n-m)!\, z^m, \qquad A_{n,m} = rH_m - (r-1)H_{n-m}, \quad H_0 = 0.$$

Define

$$R_{r,n}(z) = \frac{\sum_{m=0}^{n} w_{n,m}(z) A_{n,m}}{\sum_{m=0}^{n} w_{n,m}(z)}. \tag{7.1}$$

For positive rational z, every displayed approximant is rational. At $r = 3, z = 1$, the harmonic/binomial structure agrees with the scalar projection analyzed in [17]. The following proof treats the whole family on its own terms; it requires no unproved matrix-asymptotic hypothesis.

Theorem 7.1 (Scalar convergence for every rank)**.** *For fixed $r \geq 2$ and $z > 0$,*

$$R_{r,n}(z) = \gamma + \log z + O_{r,z}\big(n^{-(r-1)/(4r)}\big). \tag{7.2}$$

The exponent is a convenient explicit bound from a concentration argument; no optimality is asserted.

Proof. Let $x = x_n$ be the unique solution in $(0, n)$ of

$$x^r = z(n-x)^{r-1}. \tag{7.3}$$

Uniqueness follows from strict increase of the left side divided by $(n-x)^{r-1}$. This equation implies $x/n \to 0$, and then

$$x \sim z^{1/r} n^{(r-1)/r}.$$

Successive weights have ratio

$$\rho(m) = \frac{w_{n,m+1}}{w_{n,m}} = \frac{z(n-m)^{r-1}}{(m+1)^r}. \tag{7.4}$$

This is strictly decreasing as a real function on $(-1, n)$. At $x-1$ its value exceeds 1, and at x it is less than 1, by (7.3). Hence the index of a maximal weight differs from x by a bounded amount.

Put $s = x^{3/4}$. For all large n, the interval $[x-2s, x+2s]$ lies in $[x/2, 2x] \subset (0, n)$, and there

$$\frac{d}{dm}\log\rho(m) = -\frac{r-1}{n-m} - \frac{r}{m+1} \le -\frac{c_r}{x}$$

for a fixed $c_r > 0$. Summing log-ratios from a maximal index to either boundary $|m-x| = s+O(1)$ gives

$$\frac{w_{n,m}}{\max_j w_{n,j}} \le \exp(-c_r' s^2/x) = \exp(-c_r'\sqrt{x})$$

at those boundaries, with a smaller positive constant if necessary. Unimodality bounds every more distant weight by its nearer boundary weight. Since there are at most $n+1$ terms, the probability measure proportional to $w_{n,m}$ assigns the complement of $|m-x| \le s$ mass at most $O(n\exp(-c_r'\sqrt{x}))$.

Uniformly inside this window, $m = x(1+O(x^{-1/4}))$ and $n-m = (n-x)(1+O(x^{3/4}/n))$. The elementary estimate $H_k = \log k + \gamma + O(1/k)$ gives

$$\begin{aligned} A_{n,m} &= \gamma + r\log m - (r-1)\log(n-m) + O(1/x) \\ &= \gamma + r\log x - (r-1)\log(n-x) + O(x^{-1/4}) \\ &= \gamma + \log z + O(x^{-1/4}). \end{aligned}$$

On the complement, $|A_{n,m}| \le (2r-1)H_n = O_r(\log n)$. Its total contribution is therefore $O_{r,z}(n\log n\exp(-c_r'\sqrt{x}))$, negligible compared with $x^{-1/4}$. Averaging proves (7.2). □

7.2 What this proves about the external conjecture

Theorem 7.1 is an all-rank scalar result. It does not identify the weighted expression with the dual matrix output for every trajectory and every generic initial vector. Such an identification, together with control of all competing solution modes, would be required for Conjecture 1. The mere existence of one convergent scalar family does not supply it. Likewise, the convergence rate in (7.2) is not an irrationality measure: multiplying by a large common denominator can make its integer linear form grow.

One elementary extension of the source's proved matrix family can be recorded without this missing step.

Proposition 7.2 (Integer dilation of a convergent trajectory)**.** *Suppose a conservative matrix field has invertible transport matrices $T(n\mathbf{t})$ from the origin and*

$$c_n T(n\mathbf{t})^{-T} v \longrightarrow S$$

for a specified generic vector v and nonzero scalars c_n. For every positive integer d, the same state-isolation conclusion holds on the dilated trajectory $d\mathbf{t}$ with the same vector. In particular, the source's proved rank-three family extends to

$$(da_1, da_2; d, d, d), \qquad a_1, a_2 \le 0,$$

whenever the relevant transports are defined.

Proof. Conservativity makes the transport depend only on the endpoint. Transport to $n(d\mathbf{t})$ is therefore $T(dn\mathbf{t})$. The asserted convergence is the subsequence with indices dn of the assumed convergence. This proves the claim for every vector for which the original limit holds. □

This is a consequence of the source theorem, not a new solution of Conjecture 1. For example, a trajectory with unequal lower speeds or higher rank is not covered by merely dilating the proved family. The all-trajectory question remains unresolved in this paper.

8 What has been closed, and the remaining proof obligations

Saito's non-reversible recurrence and all-shift Fibonacci problems receive full answers reconstructed from the corpus. The Meijer extension resolves the selected analytic identity. The enlarged E-value field is completely classified, but arbitrary units remain outside that classification; likewise, the factorial-series theorem does not cover every nondecreasing $f \to \infty$.

For the original gamma/Mills goals, the remaining obligations can be stated without numerical ambiguity:

1. For γ, obtain nonzero *primitive integer* linear forms tending to zero, or independently establish the radical membership in Corollary 2.4. Neither follows from the available analytic identities alone.

2. For M, exclude the actual cubic-Pisot alternative, including its surviving three-adic trace limits $\lambda = \pm 1$. Conditional gap restrictions do not exclude a tower merely by describing it.

3. For ξ_2, connect rationality of the base to a contradiction in its prime sequence. Irrationality of a reciprocal series that is also irrational at rational bases cannot make that connection.

4. For the full CMF conjecture, control all competing modes of the dual system on every admissible trajectory. One explicitly convergent scalar expression does not identify a generic vector limit.

The register therefore records two full answers, one resolved identity component, and seven unresolved targets with partial results. RH, γ irrationality, and unconditional Mills transcendence are not proved.

8.1 Verification record and reproducibility

Separate internal proof checks covered the modular preperiods, prime-return threshold, finite-phase descent, power-growth constant, Barnes orientation, and scalar saddle. The phase argument explicitly clears rational basis denominators, including values such as $e^{a/2}$.

The exact verifier passed 9,200 lattice cases, 23,091 denominator divisibilities, and the displayed Lucas examples. The numerical Meijer evaluations at 35 decimal working precision used $(p, z) = (2, 1/2), (2, 2), (3, 7/10)$; the largest reported residual among the two main identities and the available $j = 2$ identity was below 10^{-30}. Those decimal diagnostics do not replace the analytic proofs. Appendix A supplies the exact verifier. The TEX needs no external input fragments, bibliography database, figures, or private data.

A A compact exact-arithmetic verifier

The following standard-library Python program checks the displayed finite lattice, common-denominator, and Lucas examples. Save the listing as a UTF-8 file and run it with Python 3. It requires no network access. Its finite output supplements the proofs and makes no claim to decide an infinite conjecture by testing a bounded range.

```
"""Finite diagnostics for the ten-target paper; standard Python only.
The infinite theorems are proved in the manuscript, not by this script.
"""
from math import factorial, gcd, isqrt
from random import Random
def check_seeds():
    count = 0
    for D in range(1, 17):
        for C in range(D):
            if gcd(C, D) != 1:
                continue
            for p in range(-7, 8):
                for q in range(1, 13):
                    if gcd(p, q) != 1:
                        continue
                    h = gcd(D*p-C*q, D*q)
                    assert D*D % h == 0
                    a, b = (D*p-C*q)//h, D*q//h
                    k = D*D//h
                    assert gcd(a, b) == 1
                    assert (D*a+C*b, D*b) == (k*p, k*q)
                    assert gcd(D*a+C*b, D*b) == k
                    if gcd(q, D) == 1:
                        assert h == 1
                    count += 1
    return count
def check_windows():
    rng = Random(20261105)
    count = 0
    for trial in range(50):
        f = [1]
        for n in range(1, 31):
            f.append(f[-1] + rng.randrange(0, 5))
        g = [n + f[n] for n in range(31)]
        for N in range(3, 31):
            options = [h for h in range(1, N)
                       if g[N]-g[N-h] >= N]
            if not options:
                continue
            h = min(options)
            L = factorial(g[N]) // factorial(N-h)
            for k in range(1, N+1):
                Dk = factorial(g[k]) // factorial(k)
                assert L % Dk == 0
                count += 1
    return count
def v2(n):
    n = abs(n)
    assert n != 0
    return (n & -n).bit_length()-1
def check_lucas():
    for P in (1, 3, 5):
        f, ell = P, P*P+2
        for r in range(1, 10):
            assert ell*ell-(P*P+4)*f*f == 4
            assert v2(ell+1) == r+1
            assert v2(f*ell*(ell*ell-2)-f) == r+1
            f, ell = f*ell, ell*ell-2
    assert all(971 % d for d in range(2, isqrt(971)+1))
```

```
    assert (174*174-174-1) % 971 == 0
    assert pow(174, 485, 971) == 1
    assert all(pow(174, 485//p, 971) != 1 for p in (5, 97))
    assert pow(2, 48, 485) == 1
    assert all(pow(2, 48//p, 485) != 1 for p in (2, 3))
    f, ell = 1, 3
    for r in range(1, 197):
        if r >= 4 and (r-4) % 48 == 0:
            assert (f-16) % 971 == 0
        f, ell = f*ell % 971, (ell*ell-2) % 971
    f, ell = 3, 0
    for r in range(3, 30):
        f, ell = f*ell % 7, (ell*ell-2) % 7
        assert (f+4) % 7 == 4
if __name__ == "__main__":
    print("PASS: primitive lattice cases:", check_seeds())
    print("PASS: denominator divisibilities:", check_windows())
    check_lucas()
    print("PASS: Lucas valuations, seed 971, and nonreturning seed 7.")
```

References

[1] K. Saito, *Mills' constant is irrational*, Mathematika **71** (2025), no. 3, e70027. Questions 1.6 and 1.7 are numbered as in arXiv:2404.19461v2, 8 May 2025. https://arxiv.org/abs/2404.19461v2; https://doi.org/10.1112/mtk.70027.

[2] K. Saito, *Transcendency of variants of Mills' constant*, The Ramanujan Journal **70** (2026), article 69, published 10 August 2026. https://doi.org/10.1007/s11139-026-01443-0. Theorem numbers used here refer to arXiv:2508.16068v3, 7 December 2025, especially Theorems 1.7 and 1.8. https://arxiv.org/html/2508.16068v3.

[3] K. Saito, *Intervals without primes near an iterated linear recurrence sequence*, International Journal of Number Theory **22** (2026), no. 6, 1269–1279. https://doi.org/10.1142/S1793042126500685. Problem numbers 1.1, 1.7 and 1.8 refer to arXiv:2504.14968v1, 21 April 2025. https://arxiv.org/html/2504.14968v1.

[4] T. Schneider, *Transzendenzuntersuchungen periodischer Funktionen I. Transzendenz von Potenzen*, Journal für die reine und angewandte Mathematik **172** (1934), 65–69. https://eudml.org/doc/149900.

[5] Keio, *Dyadic Lucas Prime Cascades and Composite Fibonacci Shifts*, user-supplied research corpus, manuscript [007], dated 5 September 2026. The full all-shift Lucas/Fibonacci proof was already present in this manuscript before the present reconstruction.

[6] Keio, *Additive Clocks, Non-Reversible Recurrences, and Transcendental Prime-Representing Constants*, user-supplied research corpus, manuscript [008], dated 5 September 2026. Source of the residue-surjectivity and non-reversible-clock results reconstructed here.

[7] Keio, *Arithmetic Constraints on a Hypothetically Algebraic Mills Constant: Gap Divisibility, Transcendental Exponents, and Cubic Reconstruction*, user-supplied research corpus, manuscript [010], dated 5 September 2026. Source of the conditional gap theorem.

[8] Keio, *Cubic Pisot Floors: Composite Windows, Sparse Exceptions, and Finite-Sieve Barriers*, user-supplied research corpus, manuscript [042], dated 21 October 2026. Source of the simultaneous three-adic composite-window criterion. These internal manuscripts are cited for provenance, not as independently peer-reviewed publications.

[9] S. Fischler and T. Rivoal, *The ring of values of E-functions is not a field*, arXiv:2610.16603, 2026. Author PDF dated 14 October 2026. https://www.imo.universite-paris-saclay.fr/~fischler/Ecroix.pdf.

[10] S. Fischler and T. Rivoal, *Arithmetic theory of E-operators*, Journal de l'École polytechnique—Mathématiques **3** (2016), 31–65. DOI: 10.5802/jep.28. https://www.imo.universite-paris-saclay.fr/~fischler/ateo.pdf.

[11] F. Beukers, *A refined version of the Siegel–Shidlovskii theorem*, Annals of Mathematics (2) **163** (2006), no. 1, 369–379. DOI: 10.4007/annals.2006.163.369. https://annals.math.princeton.edu/2006/163-1/p08.

[12] G. Lepetit, *Le théorème d'André–Chudnovsky–Katz "au sens large"*, North-Western European Journal of Mathematics **7** (2021), 81–149. https://nwejm.univ-lille.fr/index.php/nwejm/article/view/23. https://nwejm.univ-lille.fr/index.php/nwejm/article/download/23/20/40.

[13] NIST Digital Library of Mathematical Functions, Sections 5.7, 6.2, 16.17, and 16.21. Gamma expansions, exponential integrals, and the Meijer G integral and differential equation. https://dlmf.nist.gov/5.7; https://dlmf.nist.gov/6.2; https://dlmf.nist.gov/16.17; https://dlmf.nist.gov/16.21. Accessed 5 November 2026.

[14] T. Crmarić and V. Kovač, *On the irrationality of certain super-polynomially decaying series*, Colloquium Mathematicum **179** (2025), 55–68. https://doi.org/10.4064/cm9628-5-2025; https://arxiv.org/html/2504.18712v1.

[15] C. Lambrinoudis, *Algebraic Independence in the Affine Case of Erdős Problem 270*, manuscript and companion repository, version 1.1.0, 24 August 2026. https://github.com/clambro/erdos-270-transcendence. Cited for the documented history of the base case; the broad affine claims are not inputs to the present proofs.

[16] R. Kalisch, T. Monderer, E. Leibtag, S. Weinbaum, M. Shalyt, and I. Kaminer, *An Infinite Family of Apéry-like Recurrences for Euler's Constant*, arXiv:2610.13919v1, 9 October 2026. https://arxiv.org/html/2610.13919v1. Targets: Conjecture 1 and the analytic-identity component of Section 9.2(1).

[17] Keio and Shaitetsu, *Euler's Constant and Shifted Mills Constants: Arithmetic Structure, Reciprocal Series, and Proof-Transfer Limits*, integrated working manuscript, 5 November 2026, 37 pages. This is the preceding manuscript in the present research dialogue.

[18] Keio, *Verifiable Mathematical Reasoning Corpus*, working-manuscript repository, snapshot consulted 5 November 2026. https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus. The dated corpus entries above record provenance. They do not assert independent peer review or establish historical priority.